

Gradient Boosting

CSE427: Machine Learning

Why Do We Need Gradient Boosting?

Suppose one small decision tree cannot predict well.

$$\text{Actual value} \neq \text{Predicted value}$$

So the model makes errors.

Main Idea

Gradient Boosting builds many weak trees one after another. Each new tree tries to correct the errors made by the previous trees.

Instead of building one large tree:

One strong tree

Gradient Boosting builds:

$$\text{Small tree}_1 + \text{Small tree}_2 + \text{Small tree}_3 + \dots$$

What is Gradient Boosting?

Gradient Boosting is an ensemble learning algorithm.

Definition

Gradient Boosting combines many weak learners, usually small decision trees, in a sequential way.

At each iteration:

New tree learns the remaining error

For regression:

$$\text{Residual} = \text{Actual} - \text{Current Prediction}$$

For classification:

$$\text{Residual} = \text{Actual Label} - \text{Current Probability}$$

Important

Random Forest builds trees independently. Gradient Boosting builds trees sequentially.

Gradient Boosting: General Steps

- 1 Start with an initial prediction.
- 2 Calculate residuals or errors.
- 3 Train a small decision tree to predict the residuals.
- 4 Update the prediction using:

$$\text{New Prediction} = \text{Old Prediction} + \eta \times \text{Tree Prediction}$$

- 5 Recalculate residuals.
- 6 Repeat for many trees.

η = learning rate

Role of Learning Rate

Learning rate controls how much each new tree contributes.

Gradient Boosting for Regression

For regression, the target is numerical.

Example:

Weight = continuous target

Initial prediction is usually the average target value:

$$F_0(x) = \bar{y}$$

Residual:

$$r_i = y_i - F_0(x_i)$$

Then each new tree predicts the residuals.

$$F_1(x) = F_0(x) + \eta h_1(x)$$

$$F_2(x) = F_1(x) + \eta h_2(x)$$

$$F_M(x) = F_0(x) + \eta h_1(x) + \eta h_2(x) + \cdots + \eta h_M(x)$$

Regression Dataset

We want to predict weight.

ID	Height	Color	Gender	Weight
1	1.6	Blue	Male	88
2	1.6	Green	Female	76
3	1.5	Blue	Female	56
4	1.8	Red	Male	73
5	1.5	Green	Male	77
6	1.4	Blue	Female	57

Initial prediction:

$$F_0(x) = \frac{88 + 76 + 56 + 73 + 77 + 57}{6}$$

$$F_0(x) = \frac{427}{6} = 71.2$$

Step 1: Initial Residuals

Residual:

$$r_i = y_i - F_0(x_i)$$

Since:

$$F_0(x) = 71.2$$

ID	Weight	Initial Residual
1	88	$88 - 71.2 = 16.8$
2	76	$76 - 71.2 = 4.8$
3	56	$56 - 71.2 = -15.2$
4	73	$73 - 71.2 = 1.8$
5	77	$77 - 71.2 = 5.8$
6	57	$57 - 71.2 = -14.2$

Now build Tree 1 to predict these residuals.

How Does Tree 1 Create Leaves?

In Gradient Boosting, Tree 1 is not created randomly.

Important Idea

Tree 1 is trained as a regression tree, but its target is the residual.

Original target:

$$y_i$$

Current prediction:

$$F_0(x_i)$$

Residual:

$$r_i = y_i - F_0(x_i)$$

So Tree 1 learns:

Height, Color, Gender $\Rightarrow$ Residual

Residual Dataset for Tree 1

Initial prediction:

$$F_0(x) = 71.2$$

ID	Height	Color	Gender	Residual
1	1.6	Blue	Male	16.8
2	1.6	Green	Female	4.8
3	1.5	Blue	Female	-15.2
4	1.8	Red	Male	1.8
5	1.5	Green	Male	5.8
6	1.4	Blue	Female	-14.2

Question

Which feature split best separates these residuals?

How Does the Tree Choose a Split?

A regression tree tests possible splits.

Examples:

Gender = Female

Color = Blue

Height < 1.55

Height < 1.70

For each split, it calculates:

$$SSR = \sum_{\text{leaves}} \sum_{i \in \text{leaf}} (r_i - \bar{r}_{\text{leaf}})^2$$

Decision Rule

Choose the split with the smallest SSR.

Candidate Split: Gender

Try:

Gender = Female

Female residuals:

4.8, -15.2, -14.2

$$\bar{r}_{Female} = \frac{4.8 - 15.2 - 14.2}{3} = -8.2$$

Male residuals:

16.8, 1.8, 5.8

$$\bar{r}_{Male} = \frac{16.8 + 1.8 + 5.8}{3} = 8.13$$

SSR for Gender Split

Female group:

$$SSR_F = (4.8 + 8.2)^2 + (-15.2 + 8.2)^2 + (-14.2 + 8.2)^2$$

$$SSR_F = 13^2 + (-7)^2 + (-6)^2 = 254$$

Male group:

$$SSR_M = (16.8 - 8.13)^2 + (1.8 - 8.13)^2 + (5.8 - 8.13)^2$$

$$SSR_M = 75.17 + 40.07 + 5.43 = 120.67$$

Total:

$$SSR = 254 + 120.67 = 374.67$$

Final Teaching Point

So when we write:

Tree 1 creates these leaves

we really mean:

Correct Meaning

After fitting a small regression tree on the residuals, the best splits produce those terminal leaves.

The tree chooses leaves by minimizing:

$$SSR = \sum_{\text{leaves}} \sum_{i \in \text{leaf}} (r_i - \bar{r}_{\text{leaf}})^2$$

So,

Gradient Boosting does not manually choose leaves. The residual tree chooses them by the same regression-tree rule: minimum SSR.

Step 2: First Residual Tree

Suppose Tree 1 creates these terminal leaves:

Leaf	Examples	Tree 1 Prediction
Leaf A	ID 3, ID 6	$\frac{-15.2 + (-14.2)}{2} = -14.7$
Leaf B	ID 2	4.8
Leaf C	ID 4, ID 5	$\frac{1.8 + 5.8}{2} = 3.8$
Leaf D	ID 1	16.8

$$h_1(x) = \text{prediction from Tree 1}$$

Learning rate:

$$\eta = 0.1$$

Update:

$$F_1(x) = F_0(x) + 0.1h_1(x)$$

Step 3: Update Predictions After Tree 1

$$F_1(x) = 71.2 + 0.1h_1(x)$$

ID	Weight	Tree 1	New Prediction	New Residual
1	88	16.8	$71.2 + 1.68 = 72.88$	15.12
2	76	4.8	$71.2 + 0.48 = 71.68$	4.32
3	56	-14.7	$71.2 - 1.47 = 69.73$	-13.73
4	73	3.8	$71.2 + 0.38 = 71.58$	1.42
5	77	3.8	$71.2 + 0.38 = 71.58$	5.42
6	57	-14.7	$71.2 - 1.47 = 69.73$	-12.73

Step 4: Build Tree 2 on New Residuals

Now Tree 2 is trained on the latest residuals:

15.12, 4.32, -13.73, 1.42, 5.42, -12.73

Suppose Tree 2 produces:

Leaf	Examples	Tree 2 Prediction
Leaf A	ID 3, ID 6	$\frac{-13.73 + (-12.73)}{2} = -13.23$
Leaf B	ID 2	4.32
Leaf C	ID 4, ID 5	$\frac{1.42 + 5.42}{2} = 3.42$
Leaf D	ID 1	15.12

$$F_2(x) = F_0(x) + 0.1h_1(x) + 0.1h_2(x)$$

Step 5: Update Predictions After Tree 2

Example 1:

$$F_2(x_1) = 71.2 + 0.1(16.8) + 0.1(15.12)$$

$$F_2(x_1) = 71.2 + 1.68 + 1.512 = 74.392$$

$$r_1 = 88 - 74.392 = 13.608$$

Example 3:

$$F_2(x_3) = 71.2 + 0.1(-14.7) + 0.1(-13.23)$$

$$F_2(x_3) = 71.2 - 1.47 - 1.323 = 68.407$$

$$r_3 = 56 - 68.407 = -12.407$$

Main Pattern

Every new tree reduces the remaining residual.

Regression Final Model

After many trees:

$$F_M(x) = F_0(x) + \eta h_1(x) + \eta h_2(x) + \cdots + \eta h_M(x)$$

For this example:

$$F_2(x) = 71.2 + 0.1h_1(x) + 0.1h_2(x)$$

Stopping Conditions

Stop when:

- maximum number of trees is reached,
- residuals become very small,
- validation error stops improving.

Gradient Boosting for Classification

For classification, the target is categorical.

Example:

$$\text{Loves Troll2} \in \{\text{Yes}, \text{No}\}$$

Gradient boosting for classification works with:

log-odds

instead of direct numerical prediction.

Main Difference

Regression updates predictions directly. Classification updates log-odds, then converts them to probability.

$$p = \frac{1}{1 + e^{-\log \text{ odds}}}$$

Classification Dataset

ID	Loves Popcorn	Color	Age	Loves Troll2
1	Yes	Blue	12	Yes
2	Yes	Green	87	Yes
3	No	Blue	44	No
4	No	Red	19	No
5	Yes	Green	32	Yes
6	Yes	Blue	14	Yes

Counts:

$$Yes = 4, \quad No = 2$$

Initial log-odds:

$$F_0 = \log\left(\frac{4}{2}\right) = \log(2) = 0.69$$

Initial probability:

Classification Step 1: Initial Residuals

Encode labels:

$$\text{Yes} = 1, \quad \text{No} = 0$$

Residual:

$$r_i = y_i - p_i$$

Initial probability:

$$p_i = 0.67$$

ID	Actual y	Initial Probability	Residual $y - p$
1	1	0.67	0.33
2	1	0.67	0.33
3	0	0.67	-0.67
4	0	0.67	-0.67
5	1	0.67	0.33
6	1	0.67	0.33

Classification Step 2: Leaf Value Formula

For classification, a leaf does not simply use average residual.

Leaf value is:

$$\gamma = \frac{\sum \text{Residuals}}{\sum p(1 - p)}$$

where:

p = previous predicted probability

Why?

For classification, boosting updates log-odds. The denominator adjusts the residual using probability curvature.

Then update:

$$F_1(x) = F_0(x) + \eta\gamma$$

and convert to probability:

Classification Step 3: First Tree Leaves

Suppose Tree 1 produces these leaves:

Leaf	Examples	Residuals
Leaf A	ID 4	-0.67
Leaf B	ID 2, ID 3	0.33, -0.67
Leaf C	ID 1, ID 5, ID 6	0.33, 0.33, 0.33

Initial probability:

$$p = 0.67$$

So:

$$p(1 - p) = 0.67(1 - 0.67) = 0.2211$$

Classification Step 4: Compute Leaf Values

Leaf A:

$$\gamma_A = \frac{-0.67}{0.2211} = -3.03$$

Leaf B:

$$\gamma_B = \frac{0.33 - 0.67}{0.2211 + 0.2211}$$

$$\gamma_B = \frac{-0.34}{0.4422} = -0.77$$

Leaf C:

$$\gamma_C = \frac{0.33 + 0.33 + 0.33}{0.2211 + 0.2211 + 0.2211}$$

$$\gamma_C = \frac{0.99}{0.6633} = 1.49$$

$\gamma_A = -3.03, \quad \gamma_B = -0.77, \quad \gamma_C = 1.49$

Classification Step 5: Update Log-Odds

Let learning rate:

$$\eta = 0.8$$

For example 1, suppose it belongs to Leaf C:

$$F_1(x_1) = F_0 + 0.8(\gamma_C)$$

$$F_1(x_1) = 0.69 + 0.8(1.49)$$

$$F_1(x_1) = 0.69 + 1.192 = 1.882$$

Convert to probability:

$$p_1 = \frac{1}{1 + e^{-1.882}}$$

$$p_1 = 0.868$$

Classification Step 6: Update Residuals

For example 1:

$$y_1 = 1, \quad p_1 = 0.868$$

New residual:

$$r_1 = 1 - 0.868 = 0.132$$

For example 3, suppose it belongs to Leaf B:

$$F_1(x_3) = 0.69 + 0.8(-0.77)$$

$$F_1(x_3) = 0.69 - 0.616 = 0.074$$

$$p_3 = \frac{1}{1 + e^{-0.074}} = 0.519$$

Since:

$$y_3 = 0$$

Classification Final Model

After many trees:

$$F_M(x) = F_0(x) + \eta\gamma_1(x) + \eta\gamma_2(x) + \cdots + \eta\gamma_M(x)$$

Final probability:

$$p_M(x) = \frac{1}{1 + e^{-F_M(x)}}$$

Prediction rule:

$$p_M(x) \geq 0.5 \Rightarrow \text{Yes}$$

$$p_M(x) < 0.5 \Rightarrow \text{No}$$

Main Idea

Each new tree improves the log-odds, which improves the probability.

Regression vs Classification Gradient Boosting

Topic	Regression	Classification
Initial value	Mean target	Log-odds
Error term	Residual $y - \hat{y}$	Residual $y - p$
Tree predicts	Residuals	Residuals
Leaf output	Average residual	$\frac{\sum r}{\sum p(1-p)}$
Update	Prediction	Log-odds
Final output	Number	Probability/Class

Summary

- Gradient Boosting builds trees sequentially.
- Each tree corrects the previous model's error.
- For regression, trees predict residuals directly.
- For classification, trees update log-odds.
- Learning rate controls how much each tree contributes.
- Final model is the sum of the initial prediction and all tree contributions.

Gradient Boosting = Initial Prediction + Small Corrections + Many Trees